Shibbir Ahmed Arif

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Data Scientist with 3+ years of experience in machine learning, GenAI, and building end-to-end AI solutions. Skilled in Python, SQL, and Azure, with hands-on work in LLMs, RAG systems, fine-tuning, and vector databases like FAISS and ChromaDB. Experienced in developing scalable data pipelines, working with large and unstructured datasets, and deploying models using MLOps practices. Strong in feature engineering, experimentation, and turning model results into clear insights that guide product and business decisions.

TECHNICAL SKILLS

- **Programming & Tools:** Python, SQL, R, PySpark, JavaScript
- **Machine Learning & AI:** TensorFlow, PyTorch, LLM Fine-Tuning, RAG, Hugging Face Transformers, XGBoost, LightGBM, LSTM, Neural Networks, Deep Learning (CV & NLP), LangChain, Prompt Engineering, CrewAI, AutoGPT, LLM APIs (OpenAI, Groq)
- **Vector Databases & Retrieval:** FAISS, ChromaDB
- **Cloud & Data Engineering:** Azure ML, AWS (SageMaker, Bedrock), Databricks, SQL Server, Docker, ETL, Feature Engineering, Data pipelines, API Integration
- **Visualization & Web Frameworks:** Power BI, Tableau, Plotly, Streamlit, Flask, Seaborn, Matplotlib

EXPERIENCE

Product Data Scientist — DetaPent Inc. | NJ (Remote) | Jun 2025 – Present

- Build and optimize ML and GenAI pipelines using PyTorch, LangChain, FAISS, and API integrations, improving reliability and deployment readiness.
- Develop automated data workflows in Python and SQL, reducing manual work across teams.
- Deliver real-time dashboards (Power BI/Tableau) translating model outputs into actionable business insights.

Associate Data Scientist — Montclair State University Data Science Lab | Montclair, NJ | Jan 2024 – May 2025

- Developed LLM-powered NLP systems analyzing 10k+ MIMIC-IV records, improving clinical trial ranking accuracy by 20%.
- Implemented Llama 3.2–based evaluation pipeline, improving precision/recall by 35%.
- Reduced LLM response latency by 50% via vector search optimization (FAISS) and preprocessing improvements.

Data Science Intern — Community Food Bank of New Jersey | Hillside, NJ | Jun 2024 – Jan 2025

- Built automated ETL pipelines and ML-driven insights, improving distribution efficiency by 20%.
- Designed executive dashboards for grant allocation and operational decision-making.
- Integrated structured and unstructured data for scalable analytics workflows.

Senior Data Science Analyst — WorkTrooper | Dhaka, Bangladesh | Aug 2020 – Jul 2023

- Engineered ETL and database automation, improving data integrity by 25%.
- Built 100+ automated BI dashboards, improving decision-making by 30%.
- Optimized SQL and API-based data retrieval, improving performance by 15%.

KEY PROJECTS

End-to-End Medical QA Chatbot Using RAG, BioBERT & PySpark | GitHub | Live Demo

- **Enhanced clinical answer accuracy by 60%** by developing a medical chatbot for heart-related queries using BioBERT embeddings, MIMIC-IV data, and a RAG pipeline.
- **Improved retrieval precision by 50%** by using **ChromaDB** and **LangChain** to deliver more accurate context before generating final answers.

End-to-End Product Review Generation with Fine-Tuned GPT-2 (LoRA) | GitHub | Live Demo

- **Improved review quality by 95%** by fine-tuning GPT-2 LLM with LoRA on 5,000 Amazon reviews to generate product summaries.
- **Increased user interaction by 40%** by building a **Streamlit** app and tracking dashboard on **Hugging Face**, enabling real-time review generation and evaluation.

End-to-End Data Migration & Analytics Pipeline (Microsoft Azure) | GitHub

- **Improved analytics reliability by 40%** by designing an end-to-end **Azure ETL pipeline** (Data Factory → Data Lake Gen2 → Databricks) to migrate, clean, and standardize on-prem SQL Server data for BI and reporting workflows.
- **Reduced dashboard latency by 35%** by integrating transformed datasets into **Azure Synapse Analytics + Power BI**, enabling faster KPI tracking and actionable insights for product and business stakeholders.

Melanoma Skin Cancer Detection with CNN (TensorFlow) | GitHub

- **Achieved 94% accuracy** and boosted early detection rates by 30% by training a custom CNN model for melanoma classification.
- **Improved validation accuracy by 12%** by applying data augmentation and balancing the dataset for more reliable predictions across all cancer types.

EDUCATION

Master's in Data Science – Montclair State University | Montclair, New Jersey

Bachelor's in Computer Science and Engineering – Port City International University | Chittagong, Bangladesh

CERTIFICATIONS

- **Microsoft Career Essentials in Generative AI** – LinkedIn Learning
- **Google Data Analytics Professional Certificate** – Coursera